Tropical Cyclone Intensity Forecasting Using Multimodal Data Fusion Guided by Sequential Infrared Satellite Images

Chang-Jiang Zhang, Li-Yu Xu, Lei-Ming Ma, Xiao-Qin Lu, Peng He, Xue Wei, and Tao Wang

***Abstract*—Current deep learning methods for predicting tropical cyclone intensity are restricted by inadequate utilization of climate factors, insufficient feature fusion, and the inability to automatically adapt input data to changes in the scale of cyclone structures. To address these challenges, this article proposes a multimodal data fusion model guided by sequential infrared satellite images (IRGMF). The IRGMF model integrates a broader range of climate factors for a more comprehensive, enriched feature representation of tropical cyclone intensity. In addition, we designed a temporal attention module that allows the model to focus on critical moments in the sequential data for a more precise capture of the dynamic features of climate factors as they evolve. To enhance prediction accuracy, we also developed an attention-cropping module that adaptively crops irrelevant information outside the scale of a tropical cyclone from reanalysis data. To validate the effectiveness of the proposed model, we conducted experiments on multiple datasets, including CMA-BST, ERA5, and GridSat-B1. Experimental results show that, compared to existing methods, the proposed IRGMF model demonstrates significant advantages in predicting tropical cyclone intensity and provides superior accurate information for meteorological forecasting and decision support.**

***Index Terms*—Attention-cropping module, reanalysis data, sequential infrared satellite images, temporal attention module, tropical cyclone intensity prediction.**

I. Introduction

TROPICAL cyclones, which are extreme weather phenomena originating from tropical oceans, frequently pose a severe threat to coastal areas. They cause a significant threat to human life and property security through intense winds, heavy rainfall, and flooding [1], [2]. Thus, accurately predicting the intensity of tropical cyclones is crucial for guiding the public in taking timely, effective preventive measures and to mitigate potential damage [3].

Traditional methods for tropical cyclone intensity prediction are divided into three categories.

1) *Dynamical and Numerical Forecasting:* These methods are widely used in objective tropical cyclone forecasting, primarily relying on complex dynamical equations to simulate atmospheric motion and thermodynamics to predict a cyclone's path and intensity [4].
2) *Statistical Forecasting:* These methods primarily rely on historical meteorological data and statistical analysis techniques based on the correlations found between meteorological variables [5].
3) *Statistical-Dynamical Forecasting:* These methods integrate the strengths of both the statistical and dynamical approaches, aiming to correct data errors in dynamical models through statistical techniques and utilizing dynamical models to capture the physical processes of atmospheric motion [1].

However, the variation in tropical cyclone intensity is highly complex, influenced by atmospheric conditions, oceanic environments, topographical features, and global climate change. The interplay between these factors is intricate, making it difficult to describe with simple linear representations [6]. Traditional research methods fall short in capturing the nonlinear dynamics of tropical cyclones and struggle to comprehensively reveal their complex underlying physical mechanisms.

Recent developments in deep learning have attracted widespread attention, with many researchers dedicating their efforts to applying these techniques to solve various complex problems [7], [8]. Deep learning has demonstrated remarkable potential in meteorology, particularly in predicting tropical cyclone intensity. Currently, most deep learning methods for forecasting tropical cyclone intensity rely on satellite observations and reanalysis data to construct complex neural network models [9], [10], [11]. While these methods have achieved reasonable results, their limitations can be summarized as follows.

1) *Insufficient Utilization of Climate Factors:* When using reanalysis data, traditional deep learning methods [12],

Received 30 March 2025; revised 7 August 2025; accepted 21 September 2025. Date of publication 1 October 2025; date of current version 9 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42575003 and Grant 42075140; and in part by the Project of Commonwealth Techniques and Application Research of Zhejiang Province, China, under Grant TGG24D050001. *(Chang-Jiang Zhang and Li-Yu Xu contributed equally to this work.) (Corresponding authors: Chang-Jiang Zhang; Lei-Ming Ma.)*

Chang-Jiang Zhang is with the School of Artificial Intelligence and Zhejiang Provincial Key Laboratory of Plant Evolutionary Ecology and Conservation, School of Life Sciences, Taizhou University, Taizhou 318000, China (e-mail: zcj74922@zjnu.edu.cn).

Li-Yu Xu is with the College of Physics and Electronic Information Engineering, Zhejiang Normal University, Jinhua 321004, China (e-mail: 2297990871@qq.com).

Lei-Ming Ma and Xiao-Qin Lu are with Shanghai Typhoon Institute, China Meteorological Administration, Shanghai 200030, China (e-mail: malm@typhoon.org.cn; luxq@typhoon.org.cn).

Peng He and Tao Wang are with the School of Artificial Intelligence, Taizhou University, Taizhou 318000, China (e-mail: 260269139@qq.com; 6799084@qq.com).

Xue Wei is with Taizhou Meteorological Bureau, Taizhou 317000, China (e-mail: w381594874@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3616026

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

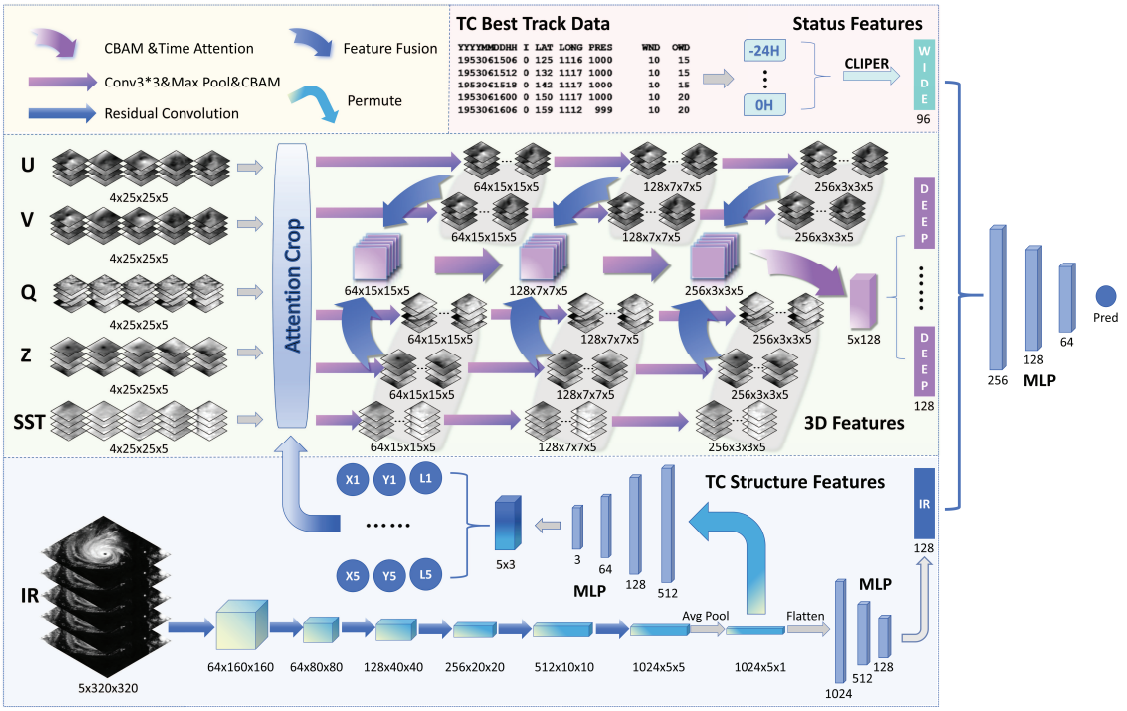


Fig. 1. Framework of our proposed IRGMF module.

[13], [14] often focus solely on the U and V components of wind as inputs, which primarily reflect the atmospheric flow within a tropical cyclone system. Despite being reasonable, this approach overlooks equally important climate factors in the dataset that may play indispensable roles.

2) *Inadequate Feature Fusion:* After encoding meteorological variables, most methods [14], [15], [16], [17] neglect the interaction and fusion of the features in the time dimension, leading to the model's inability to accurately capture and reflect temporal variation patterns.
3) *Inability to Automatically Adapt the Input Data Range to Changes in Scale:* Most methods [12], [13], [14], [15], [16], [17] typically crop reanalysis data into fixed-sized images centered on the tropical cyclone, which are fed into the model. However, the scale varies over time with changing environmental conditions. Fixed-range input data include considerable irrelevant information, which affects the model's performance.

To overcome this, this study developed an innovative multimodal data fusion model guided by sequential infrared satellite images (IRGMF), as shown in Fig. 1. In contrast to comparable models in the IRGMF framework, our model obtains a comprehensive, enriched feature representation of tropical cyclone intensity by integrating a broad range of climate factors. We have also designed a temporal attention module that enables the model to focus on critical time points within the sequential data, thereby more accurately capturing the dynamic features of climate factors as they evolve. To improve accuracy, we developed an attention-cropping module that eliminates irrelevant information outside the cyclone. The core contributions of this study are as follows.

1) In addition to adopting the commonly used U and V components of wind (U and V) used by most deep learning methods, we integrated a variety of climate factors, including sea surface temperature (SST), specific humidity (Q), and geopotential (Z), to construct a comprehensive, detailed dataset for forecasting tropical cyclone intensity. This provides robust data support for subsequent model training, enriches the depth and hierarchy of feature representation, and significantly improves the accuracy of tropical cyclone intensity predictions.
2) We propose a temporal attention module so the model can focus on critical time points within the sequential data. This mechanism enables our model to more accurately captures the dynamic features of climate factors related to tropical cyclones as they evolve, effectively identifying the key time points crucial for intensity forecasting.
3) We also introduce an attention-cropping module to precisely crop reanalysis data to remove irrelevant information outside the scale of the cyclone. Cropping enhances the model's ability to recognize tropical cyclone features, further improving the prediction accuracy and computational efficiency of the forecasting model.

The remainder of this article is organized as follows. Section II reviews related research on forecasting tropical cyclone intensity. Section III presents the IRGMF prediction model proposed in this study, and Section IV validates the effectiveness of our model through a series of experiments. Finally, Section V concludes this article.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

A. Traditional Tropical Cyclone Intensity Forecasting Methods

Traditional methods for forecasting tropical cyclone intensity typically use numerical forecasting. This approach has been adopted by such meteorological organizations as the China Meteorological Administration (CMA) in its Global/Regional Assimilation and Prediction System [18], the European Centre for Medium-Range Weather Forecasts (ECMWF) in its Integrated Forecast System (IFS) [19], and the National Centers for Environmental Prediction (NCEP) in its Global Forecast System (GFS) [20]. These systems provide detailed global climate forecasts through complex physical models and numerical computations.

The CLImatology and PERsistence (CLIPER) method of statistical forecasting was proposed by Jarvinen and Neumann [21], opened a new path for predicting tropical cyclone intensity through statistical regression. Building on this, Chu [22] developed a statistical regression model to forecast tropical cyclone intensity in the Northwest Pacific. Using statistical-dynamical methods, DeMaria and Kaplan [23] constructed an equation relating predictor factors and tropical cyclone intensity through multiple linear regression analyses, where determining atmospheric predictors relies on numerical forecasting models. Zhang et al. [24] further explored the possibility of incorporating large-scale climate change features into the forecasting factors of statistical-dynamical methods.

However, traditional methods have proven insufficient in capturing the nonlinear dynamic characteristics of tropical cyclones, making it difficult to fully reveal their underlying complex physical mechanisms.

B. Deep Learning Methods for Tropical Cyclone Intensity Forecasting

Many researchers have recently applied deep learning techniques to tropical cyclone intensity forecasting. For example, Tong et al. [25] introduced a convolutional long short-term memory (ConvLSTM) network that effectively extracts spatiotemporal features by combining the spatial feature extraction capabilities of convolutional neural networks (CNNs) with the sequential modeling ability of long short-term memory (LSTM) networks. However, despite its advantages in capturing spatiotemporal dynamic changes, the model falls short of fully analyzing the intricate interactions among multiple variables. Huang et al. [26] proposed MMSTN, a probabilistic prediction model that generates multiple possible trajectory and intensity tendencies of tropical cyclones, providing more informative guidance for forecasters. While MMSTN performs well in short-term forecasting and effectively integrates multimodal inputs, it lacks diverse data sources such as infrared satellite imagery, limiting its ability to capture fine-scale meteorological features. In addition, it does not include spatial interpretability mechanisms such as attention visualization.

Studies have been limited in selecting input data for the models. For instance, Xu et al. [14] used only the U and V components of wind and observational data as model inputs without adequately considering the influence of other important climate factors in the reanalysis dataset. Huang et al. [27] proposed a method that can utilize heterogeneous meteorological data efficiently and mine environmental factors. However, within the reanalysis data, the method incorporated only Z. Park et al. [28] explored long-term predictions by integrating physical values from unified model data, which are outputs from a real-time-capable numerical weather prediction model, with the power of data-driven methods. However, their model inputs were limited to Z, the wind vector, and observational data. This restricted the model's ability to comprehensively account for complex environmental factors. In terms of model architecture, Li et al. [13] adopted an innovative complex spatial attention mechanism to enhance a model's spatial awareness. However, they did not effectively incorporate interactions between features on the temporal channel and failed to fully explore the dynamic dependencies inherent in sequential data. Wang et al. [15] took a different approach by enhancing the original dataset with heatmaps generated after model training, but this method introduced irrelevant information outside the scale of the tropical cyclone and negatively affected prediction accuracy. To address these problems, this article proposes a novel tropical cyclone intensity forecasting model, IRGMF, which combines deep learning with multimodal satellite data.

III. Methodology

A. Problem Definition

The essence of tropical cyclone intensity forecasting based on deep learning is a spatiotemporal regression problem. Given the historical sequence data $\{\chi_{t-\alpha i}, \chi_{t-(\alpha-1)i}, \ldots, \chi_t\}$ of a tropical cyclone up to the current time t with a length of $\alpha + 1$, where i is the time interval between adjacent data points in the sequence, our objective is to forecast the intensity of a tropical cyclone at a particular point in time, denoted $t + \beta i$. The prediction process can be formally expressed as follows:

$$\hat{Y}_{t+\beta i} = \arg\max p\left(Y_{t+\beta i} | f\left(\chi_{t-\alpha i}, \chi_{t-(\alpha-1)i}, \ldots, \chi_t\right)\right). \quad (1)$$

Per the work of Li et al. [13], we set α and β to 4, and the length i to 6, using a 24-h historical sequence of tropical cyclone data to predict the intensity of a tropical cyclone for the following 24 h.

B. Overview

The network architecture proposed in this study is shown in Fig. 1 and consists of the following core elements.

1) Tropical Cyclone State Feature Extraction Module: This module uses the CLIPER method [29], [30] to process the historical observational data of tropical cyclones. The method can generate high-quality features that align with the long-term statistical patterns and short-term variational trends of tropical cyclones, thereby optimizing feature representation for greater accuracy and reliability.

2) Tropical Cyclone 3-D Feature Extraction Module: This module encodes the reanalysis data of a tropical cyclone using multiple convolutions. During this process, key operations such as feature fusion and temporal attention are applied to explore and optimize the accuracy of extracting 3-D features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TROPICAL CYCLONE STATE FEATURES

ID	Features
1-10	$lat_0, lat_{-6}, lat_{-12}, lat_{-18}, lat_{-24}, long_0, long_{-6}, long_{-12}, long_{-18}, long_{-24}$
11-20	$wind_0, wind_{-6}, wind_{-12}, wind_{-18}, wind_{-24}, pres_0, pres_{-6}, pres_{-12}, pres_{-18}, pres_{-24}$
21-24	$lat_0 - lat_{-6}, lat_{-6} - lat_{-12}, lat_{-12} - lat_{-18}, lat_{-18} - lat_{-24}$
25-28	$long_0 - long_{-6}, long_{-6} - long_{-12}, long_{-12} - long_{-18}, long_{-18} - long_{-24}$
29-32	$wind_0 - wind_{-6}, wind_{-6} - wind_{-12}, wind_{-12} - wind_{-18}, wind_{-18} - wind_{-24}$
33-36	$pres_0 - pres_{-6}, pres_{-6} - pres_{-12}, pres_{-12} - pres_{-18}, pres_{-18} - pres_{-24}$
37-41	$\cos(lat_0), \cos(lat_{-6}), \cos(lat_{-12}), \cos(lat_{-18}), \cos(lat_{-24})$
42-45	$\cos(lat_0) - \cos(lat_{-6}), \cos(lat_{-6}) - \cos(lat_{-12}), \cos(lat_{-12}) - \cos(lat_{-18}), \cos(lat_{-18}) - \cos(lat_{-24})$
46-50	$\log(wind_0), \log(wind_{-6}), \log(wind_{-12}), \log(wind_{-18}), \log(wind_{-24})$
51-55	$\log(pres_0), \log(pres_{-6}), \log(pres_{-12}), \log(pres_{-18}), \log(pres_{-24})$
56-65	$(wind_0)^2, (wind_{-6})^2, (wind_{-12})^2, (wind_{-18})^2, (wind_{-24})^2, (wind_0)^3, (wind_{-6})^3, (wind_{-12})^3, (wind_{-18})^3, (wind_{-24})^3$
66-70	$pres_0/wind_0, pres_{-6}/wind_{-6}, pres_{-12}/wind_{-12}, pres_{-18}/wind_{-18}, pres_{-24}/wind_{-24}$
71-75	$wind_0/pres_0, wind_{-6}/pres_{-6}, wind_{-12}/pres_{-12}, wind_{-18}/pres_{-18}, wind_{-24}/pres_{-24}$
76-80	$pres_0/(wind_0)^2, pres_{-6}/(wind_{-6})^2, pres_{-12}/(wind_{-12})^2, pres_{-18}/(wind_{-18})^2, pres_{-24}/(wind_{-24})^2$
81-85	$(wind_0)^2/pres_0, (wind_{-6})^2/pres_{-6}, (wind_{-12})^2/pres_{-12}, (wind_{-18})^2/pres_{-18}, (wind_{-24})^2/pres_{-24}$
86-90	$pres_0/(wind_0)^3, pres_{-6}/(wind_{-6})^3, pres_{-12}/(wind_{-12})^3, pres_{-18}/(wind_{-18})^3, pres_{-24}/(wind_{-24})^3$
91-95	$(wind_0)^3/pres_0, (wind_{-6})^3/pres_{-6}, (wind_{-12})^3/pres_{-12}, (wind_{-18})^3/pres_{-18}, (wind_{-24})^3/pres_{-24}$
96	*Month*

3) Tropical Cyclone Structural Scale Feature Extraction Module: The primary function of this module is to apply multiple convolutions to infrared satellite images to accurately capture the structural features and scale of a tropical cyclone. Subsequently, the module uses the extracted features to crop the reanalysis data, eliminating irrelevant information outside the scale of the cyclone.

The state, 3-D, and structural scale features extracted by the modules are integrated and fed to a four-layer fully connected neural network to obtain the final tropical cyclone intensity forecast.

C. Tropical Cyclone State Feature Extraction Module

The input data used by this module are from the Tropical Cyclone Best Track (CMA-BST) dataset [31]. This dataset contains the cyclone's identification number, name, the latitude and longitude coordinates of its center, the 2-min average maximum wind speed near its center, the central minimum pressure, and the 2-min average wind speed. We selected four key factors from the dataset: the longitude and latitude of the cyclone center, the central minimum pressure, and the 2-min average maximum wind speed near the center. These factors represent the basic state characteristics of a cyclone and are essential for forecasting intensity changes.

We multiplied the longitude and latitude values by 0.1 and excluded samples in the dataset containing zero values to avoid division-by-zero errors. However, treating these climate factors directly as state features neglects their interactions. To generate state features that adhere to both the long-term statistical patterns and short-term variational trends of tropical cyclones, we adopted the CLIPER method, which has been widely applied in tropical cyclone tracking and intensity forecasting. By using differential terms, quadratic terms, and sine functions, we constructed 96 predictive factors (see Table I). These factors form the cyclone state features that describe its current state and reveal its future development.

D. Tropical Cyclone 3-D Feature Extraction Module

The reanalysis data used in this module are sourced from the fifth-generation atmospheric reanalysis of the global climate (ERA5) [32], which is widely applied in global atmospheric reanalysis studies. This dataset contains a wealth of climate factors, providing researchers with valuable climate data, such as wind speed, humidity, and temperature, which strongly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

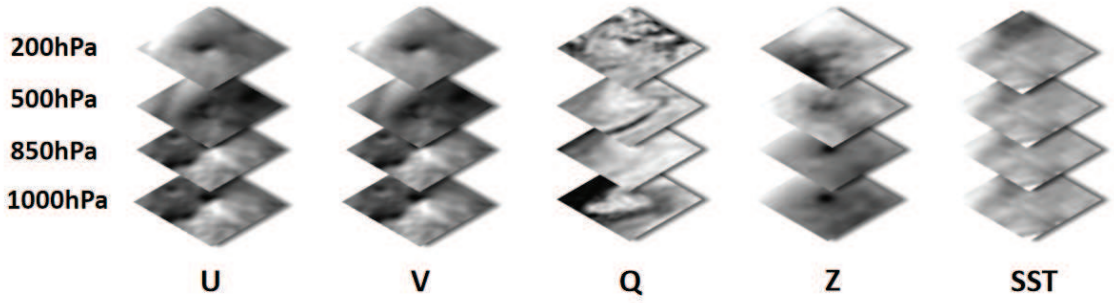


Fig. 2. Climate factors at four isobaric levels.

TABLE II
CLIMATE FACTORS USED IN THE MODEL

Type	Full Name	Abbreviation
Upper-air factors	Geopotential	Z
	Specific humidity	Q
	U component of wind	U
	V component of wind	V
Surface factors	Sea Surface Temperature	SST

support a deeper understanding and analysis of the global climate system.

However, traditional deep learning methods often rely solely on the U and V wind components for inputs when utilizing reanalysis data. These components primarily represent airflow dynamics within a tropical cyclone system. While this approach is reasonable, it inevitably overlooks other critical climate factors in the dataset that are also indispensable for accurate tropical cyclone intensity forecasting.

In light of this, this study selects several key climate factors for in-depth analysis, including Z, Q, U, V, and SST. Z represents the potential energy of an air parcel influenced by altitude and gravity in the atmosphere. Its variations intuitively reflect the distribution and dynamic energy changes within the tropical cyclone system [33]. Q, which quantifies the water vapor content in the air, is a crucial physical parameter. For tropical cyclones, water vapor is a fundamental component of the energy cycle and a determining factor in their development and intensity changes. High-humidity facilitates latent heat release through condensation and increases intensity [34]. SST is an indicator of ocean surface temperature and directly affects the rate of water vapor evaporation and the intensity of the energy supply, playing a critical role in the formation, development, and intensification of tropical cyclones [35].

In summary, the climate factors U, V, Q, Z, and SST collectively provide a comprehensive description of a tropical cyclone system's physical and dynamic characteristics (see Table II), offering richer, more complete, and more accurate information to support tropical cyclone intensity forecasting.

We selected data from four specific isobaric levels as inputs to the model, including 200, 500, 850, and 1000 hPa (see Fig. 2). Since SST data in the reanalysis data are available at only a single level, we replicated the SST data three times to maintain consistency with other climate factors. This approach allows for direct comparison and analysis of SST alongside other climate variables at the same isobaric levels. The spatial resolution of the reanalysis data is 0.25°. To ensure computational efficiency, we extracted data within a 100 × 100 grid centered on the cyclone and subsequently downscaled them to a 25 × 25 pixel resolution.

Next, we employed convolutional layers with shared weights to extract fundamental features from each climate factor. This approach ensures that the extracted features are representative and robust by capturing common distinctive characteristics across different climate variables. To further enhance the model's feature extraction capability, we incorporated the convolutional block attention module (CBAM) [36], an attention mechanism that improves feature representation by automatically identifying and focusing on key regions within the input data. Including CBAM allows the model to more accurately pinpoint regions that significantly influence wind speed variations, thereby improving the accuracy of forecasting tropical cyclone intensity.

After each convolution operation, we performed feature fusion to integrate features from various levels for a richer, more comprehensive representation. This way, the model preserves low-level meteorological details and captures high-level abstract features, achieving multilevel and multidimensional information integration during deep learning. The mathematical expression for feature fusion is shown in the following equation:

$$h_{C_{i+1}} = \alpha_{C_i} h_{C_i} + \sum_{k=1}^{5} \alpha_{M_i^K} h_{M_i^K} \tag{2}$$

where h_{C_i} represents the fused features from the previous layer (set to 0 for the first layer of fusion), α_{C_i} is the weight assigned to the fused features from the previous layer, and $\alpha_{M_i^K}$ is the weight assigned to each meteorological feature $h_{M_i^K}$ after convolution, with the constraint $\alpha_{C_i} + \sum_{k=1}^{5} \alpha_{M_i^K} = 1$. The features resulting from three layers of convolution and fusion are then further processed through our designed temporal attention module (see Fig. 3) to enhance the interaction and integration of features along the temporal dimension, thereby more deeply revealing the spatiotemporal evolution patterns of tropical cyclones. Specifically, the design of the temporal attention module enables the model to focus strongly on key time points within the sequential data, more accurately

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

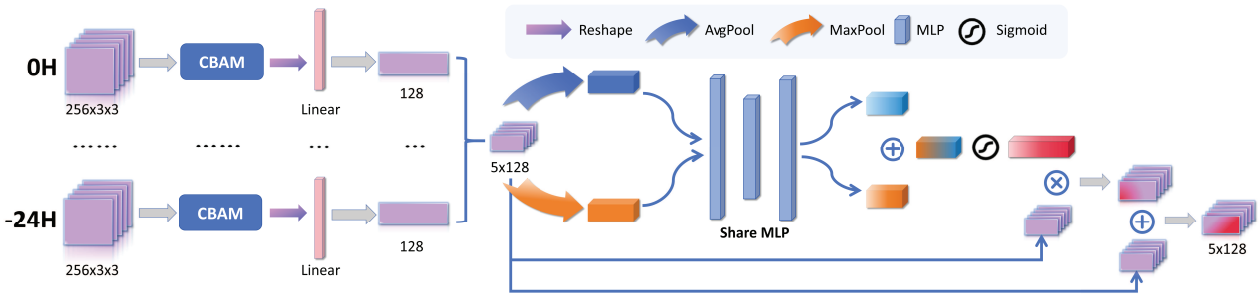


Fig. 3. Temporal attention module.

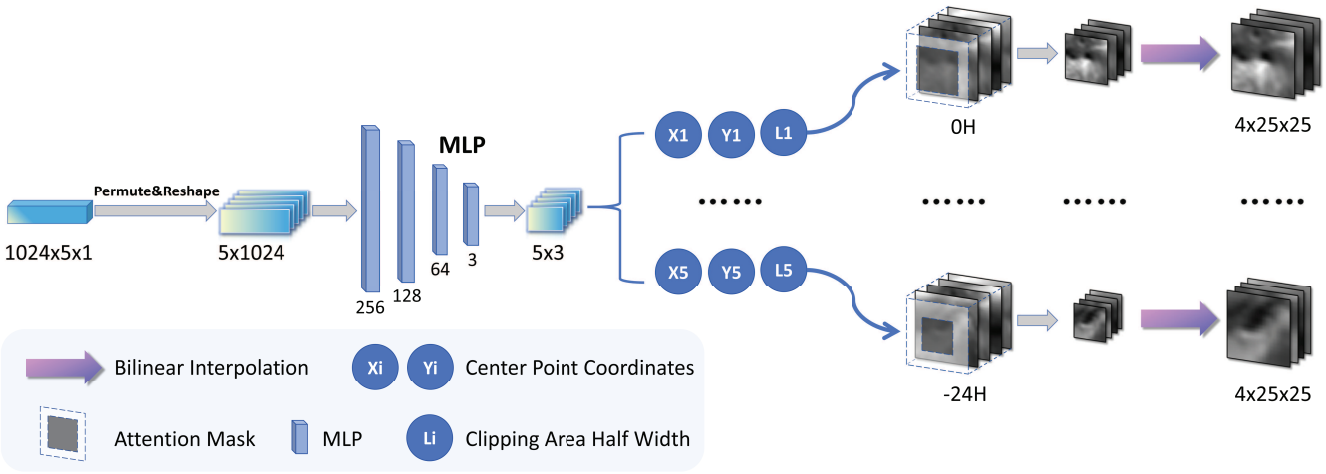


Fig. 4. Attention-cropping module.

capturing the dynamic features of climate factors associated with tropical cyclones as they evolve. The module first applies the CBAM attention mechanism to enhance features extracted from five time points and then employs a reshape function to adjust their dimensions. Next, the features at each time step are transformed into a 128-dimension representation after processing through a linear layer. Subsequently, the adjusted features are integrated, and the most representative information is extracted using a combination of max pooling and average pooling. After pooling, the features are fed into a multilayer perceptron (MLP) with shared weights for further processing; the MLP extracts more abstract and representative features. Upon completion, the two sets of features output by the MLP are summed and then passed through a sigmoid function to generate temporal attention weights that reflect the importance of each time point, enabling the model to reweight the original fused features accordingly and thereby concentrate on the critical points. After multiplying the attention weights with the original features, a skip connection is performed. This operation combines low-level and high-level features, generating a richer and more comprehensive representation. This allows the model to more effectively capture the critical dynamic characteristics within the sequential data related to tropical cyclones. Through the multilevel feature extraction and fusion mechanisms, along with the optimized application of the temporal attention module, our model can extract a more comprehensive and enriched set of 3-D features of tropical cyclones.

E. Tropical Cyclone Structural Scale Feature Extraction Module

The input data for this module are obtained from the GridSat-B1 dataset [37], which provides users with long time series and high-resolution global infrared climate data. Although infrared climate data are not fully exploited by conventional deep learning methods, this demonstrates the unique advantages in revealing the cloud structures of tropical cyclones. Specifically, these data can accurately capture key features of tropical cyclones, such as the central dense cloud region, spiral rainbands, and the central cold cloud shield, which are crucial for assessing a cyclone's intensity and developmental stage. For instance, the presence of a central dense cloud region is often an indicator of high cyclone intensity, while the clarity and curvature of the spiral rainbands directly reflect the cyclone's strength and rotational speed. In this study, we first cropped 320 × 320 pixel images centered on tropical cyclones from the original infrared satellite imagery and then overlaid the images from five time points (−24, −18, −12, −6, and and 0 h) to enrich the spatiotemporal information. Subsequently, these composite images are fed to a multilevel residual CNN (see Table III) for feature extraction,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

TABLE III
INFRARED CLOUD IMAGE FEATURE EXTRACTION MODULE

Layer Name	Operation	Operation Setting	Output size
Feature Extraction	Conv&BN&ReLU MaxPool	Input Channels=5;Output Channels=64; kernel=7x7;Stride=2;	64x160x160
Residual Convs 1	Conv&BN&ReLU& Conv&BN&ReLU& Conv&BN	$\begin{bmatrix}\text{kernal}=3x3;\text{Output Channel}=64\\ \text{kernal}=3x3;\text{Output Channel}=64\end{bmatrix}x3$	64x80x80
Residual Convs 2		$\begin{bmatrix}\text{kernal}=3x3;\text{Output Channel}=128\\ \text{kernal}=3x3;\text{Output Channel}=128\end{bmatrix}x4$	128x40x40
Residual Convs 3		$\begin{bmatrix}\text{kernal}=3x3;\text{Output Channel}=256\\ \text{kernal}=3x3;\text{Output Channel}=256\end{bmatrix}x6$	256x20x20
Residual Convs 4		$\begin{bmatrix}\text{kernal}=3x3;\text{Output Channel}=512\\ \text{kernal}=3x3;\text{Output Channel}=512\end{bmatrix}x3$	512x10x10
Residual Convs 5		$\begin{bmatrix}\text{kernal}=3x3;\text{Output Channel}=1024\\ \text{kernal}=3x3;\text{Output Channel}=1024\end{bmatrix}x3$	1024x5x5
Representation Generator	AdaptiveAvgPool2d	Output Size = (5,1)	1024x5x1

and the network's outputs are downsampled using an average pooling layer. After feature extraction and dimensionality reduction, the data are divided into two branches for further analysis. In the first branch, the data are processed through three fully connected layers to extract the structural scale features of the cyclone, which are critical for understanding their internal structure and dynamic evolution. In the second branch, the data are passed through an attention-cropping module (see Fig. 4) to determine the scale of the cyclone at each point and to selectively crop the climate factor images, thereby effectively eliminating irrelevant information that is out of scale. The attention cropping enhances the model's ability to recognize and capture key features, providing more precise data support for subsequent analysis and forecasting. The execution of the attention-cropping module can be summarized as follows. First, a permute instruction is applied to rearrange the dimensions of the input data with the time dimension at the forefront. Next, the data are sequentially fed along the time dimension into a four-layer fully connected network that outputs the cropping center coordinates (X, Y) and the half-width (L) of the cropping region for each time point. An attention mask is generated based on these outputs to identify the key regions within the data. Next, using the Hadamard product, we fuse the original climate factor data with the attention mask. Finally, to ensure that the cropped images meet the input requirements of the tropical cyclone 3-D feature extraction module, bilinear interpolation is employed to upscale the images.

IV. EXPERIMENT AND ANALYSIS

To validate the reliability of our model, we conducted extensive experiments on datasets, including CMA-BST, ERA5, and GridSat-B1. In this section, we first describe the data used and the evaluation criteria employed. Next, we provide detailed information on the experimental design. Finally, an analysis of the experimental results demonstrates the effectiveness and practicality of the proposed model.

A. Datasets

In this study, we utilized three datasets: 1) CMA-BST dataset from CMA (https://tcdata.typhoon.org.cn), which records the positions and intensity variations of tropical cyclones in the Northwestern Pacific; 2) the ERA5 reanalysis dataset from ECMWF (https://cds.climate.copernicus.eu), which comprehensively integrates global observational data and, through rigorous quality control and data assimilation, provides detailed and reliable meteorological and surface parameters; and 3) the GridSat-B1 dataset from the National Oceanic and Atmospheric Administration (NOAA) (https://www.ncei.noaa.gov/products/gridded-geostationary-brightness-temperature), which leverages global infrared measurements from geostationary satellites to establish a high-quality climate data system that robustly supports scientific analysis. We aim to fuse the information from these three datasets to enhance the diversity and complementarity of the model's input data and improve the accuracy of tropical cyclone intensity forecasting. Given that the temporal resolutions of the datasets differ (with CMA-BST at 6-h interval and ERA5 and GridSat-B1 at 1-h interval), we adjusted the temporal resolutions of ERA5 and GridSat-B1 to match that of CMA-BST. We partitioned the data by year into training, validation, and test sets to ensure fairness and accuracy in evaluating the model's predictive performance. Specifically, data from 1981 to 2010 were used for training,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

data from 2011 to 2014 for validation, and data from 2015 to 2018 for testing. We excluded tropical cyclone samples shorter than 48 h during preprocessing and removed samples with missing time point data. The final dataset consisted of 18 363 samples in the training set, 2379 in the validation set, and 2758 in the test set.

B. Evaluation Criteria and Implementation Details

We employed the mean absolute error (MAE) as the primary evaluation metric to precisely quantify the deviation between the model's predicted intensity and the best track intensity of tropical cyclones. This computation process can be expressed as

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n} |\hat{y}_i - y_i|}{n} \tag{3}$$

where n represents the total number of samples, $\hat{y}_i$ denotes the model's predicted intensity for the i sample, and y_i is the best track intensity of the corresponding sample. Both the model's predicted intensity and the best track intensity refer to the 2-min mean maximum sustained wind speed near the tropical cyclone center.

The model was trained over 64 epochs, with a batch size of 96. We employed the Adam optimization algorithm for gradient descent, setting the initial learning rate to 0.0007 and adjusting it using cosine annealing. Our model was implemented and executed entirely within the PyTorch framework. All training and validation procedures were performed on a machine equipped with an Intel i9-12900 processor and an NVIDIA RTX 3090 GPU.

C. Performance

1) Comparison Methods: To show the effectiveness of our proposed method, we compared the proposed method with representative current methods.

a) Subjective forecasting methods: We analyzed the tropical cyclone forecasts produced by three authoritative meteorological agencies: CMA, the Japan Meteorological Agency (JMA), and the United States Joint Typhoon Warning Center (JTWC).

b) Global numerical forecasting methods: We considered two global numerical weather prediction models: IFS from ECMWF and GFS from NCEP. These models are widely recognized for their advanced computational techniques and extensive meteorological datasets, enabling comprehensive global weather predictions.

c) Regional numerical forecasting methods: We included three regional numerical forecasting models in our analysis: CMA-TRAMS and CMA-TYM, developed by the CMA, and BoM-ACCESS-TC, developed by the Australian Bureau of Meteorology. These models are designed with higher spatial resolution and are specifically tailored for tropical cyclone forecasting in their respective regions.

d) Statistical forecasting methods: We examined the classical CLIPER statistical forecasting model [22], which demonstrates effective predictive capability for tropical cyclone intensity variations through a set of carefully selected statistical variables. Furthermore, the CLIPER+EnvAvg model extends CLIPER by incorporating additional environmental factors derived from reanalysis data to enhance prediction accuracy.

e) Machine learning methods: We specifically investigated XGBoost, a highly efficient machine learning algorithm with outstanding nonlinear fitting capabilities that provides more precise tropical cyclone intensity predictions based on CLIPER features. The XGBoost+EnvAvg method further improves prediction comprehensiveness and accuracy by integrating environmental factors from reanalysis data.

f) Deep learning methods: The CLIPER+ConvLSTM model innovatively combines CLIPER's precise transformation of observational data with ConvLSTM's spatiotemporal modeling capabilities, enabling a deeper analysis of dynamic variations in reanalysis data [29]. The SAF-Net model incorporates a spatial attention module to automatically identify high-response wind speed regions, further improving forecast accuracy [14]. In addition, the MAF-Net model employs a complex convolutional encoding module that utilizes a unique complex representation to capture wind field orthogonality and multiscale features, thereby providing a more refined numerical foundation for tropical cyclone prediction [13].

2) Performance Comparison: We utilized MAE to compare our proposed method with representative current methods. The experimental results are shown in Table IV. By analyzing the data in the table, we derived the following conclusions.

a) Subjective forecasting methods: CMA and JMA achieved favorable forecasting performance on the test set in different years. Specifically, CMA exhibited lower prediction errors in 2015 and 2016, while JMA outperformed in 2017 and 2018. Despite these results, our proposed IRGMF model consistently outperformed the best numerical prediction method from 2015 to 2017 in terms of lower forecasting errors and matched its performance in 2018. Notably, in 2016 and 2017, the IRGMF model significantly reduced the error by 14.5% and 31.0%, respectively.

b) Global and regional numerical forecasting methods: Among numerical forecasting methods, ECMWF-IFS achieved the lowest forecast error in 2015, NCEP-GFS performed best in 2016 and 2018, and CMA-TYM showed the lowest error in 2017. In comparison, IRGMF achieves a performance improvement of 0.9–2.06 m/s over these methods.

c) Statistical and machine learning methods: Among statistical and machine learning methods, the XGBoost model demonstrated outstanding performance. By leveraging CLIPER features, it effectively utilized its superior nonlinear fitting capabilities. In 2017, the XGBoost model reduced the prediction error by 18.6% compared to the best numerical prediction method. The XGBoost+EnvAvg model further improved on XGBoost by incorporating additional environmental factors from reanalysis data, reducing the four-year average intensity prediction error. However, our proposed IRGMF model consistently achieved lower intensity prediction errors than XGBoost and its improved variant across all years from 2015 to 2018.

d) Deep learning methods: The CLIPER+ConvLSTM and SAF-Net models were improved significantly by integrating tropical cyclone observational data with reanalysis

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF METHODS BY MAE FOR TROPICAL CYCLONE INTENSITY FORECASTING FOR LEAD TIMES OF 24 H (UNIT: M/S)

Type	Methods	2015	2016	2017	2018	Avg
Subjective Forecasting	CMA	4.26	5.10	5.63	4.00	4.75
	JMA	5.08	5.30	5.05	3.90	4.83
	JTWC	4.88	5.40	5.23	4.70	5.05
Global Numerical Forecasting	ECMWF-IFS	5.44	9.22	6.40	6.90	6.99
	NCEP-GFS	6.42	6.42	5.92	4.80	5.89
Regional Numerical Forecasting	CMA-TRAMS[40-42]	8.90	8.00	6.50	6.40	7.45
	CMA-TYM[40-42]	—	—	4.60	5.50	5.05
	BoM-ACCESS-TC[40-42]	7.40	6.80	5.60	6.90	6.67
Statistical Forecasting	CLIPER[22]	5.20	6.17	4.85	5.45	5.42
	CLIPER+EnvAvg	5.16	6.12	4.82	5.32	5.36
Machine Learning	XGBoost[43]	4.78	5.43	4.11	4.28	4.65
	XGBoost+EnvAvg[43]	4.69	5.39	4.13	4.12	4.58
Deep Learning	CNN-based[44]	6.79	7.30	6.14	6.95	6.80
	CLIPER+ConvLSTM[29]	4.77	5.34	4.03	4.25	4.60
	SAF-Net[14]	4.54	4.78	3.95	3.94	4.30
	MAF-Net[13]	4.34	4.97	3.82	**3.71**	4.19
	IRGMF	**4.00**	**4.36**	**3.48**	3.90	**3.93**

data. These multimodal approaches demonstrated superior accuracy compared to traditional CNN methods that rely solely on reanalysis data. This result highlights the advantages of multimodal data utilization, as it provides richer and more comprehensive information, effectively mitigating potential biases caused by reliance on a single data modality. The MAF-Net model further introduced a complex convolutional encoding module to represent wind fields, enabling more precise multiscale feature extraction during encoding. Due to the fine-grained feature representation, MAF-Net achieved the lowest forecasting error of 3.71 m/s in 2018, while our IRGMF model yielded a prediction error of 3.90 m/s. Except for MAF-Net, our model outperformed all other methods regarding prediction accuracy. Compared to MAF-Net, the IRGMF model reduced prediction errors by 7.8%, 12.2%, and 8.9% in 2015, 2016, and 2017, respectively. Furthermore, by averaging the four-year test set data, our model improved overall performance significantly, reducing the mean error by 6.2%. These results validate our model's effectiveness and underscore the potential of using infrared satellite images to improve the accuracy of forecasting tropical cyclone intensity.

3) Ablation Study: We conducted ablation experiments to further validate and explore the effectiveness of different climate factors and the attention module in our model.

a) Different climate factors: In this experiment, we first established a baseline model using only the U and V and tropical cyclone observational data (O). Then, we incrementally included Q, Z, SST, and IR. By progressively increasing the diversity and complexity of the input data, we assessed their impact on model performance. Table V shows a clear trend: the four-year average error on the test set gradually decreases as more data types are incorporated. Specifically, when the model relies solely on the U, V, and O, the tropical cyclone intensity forecast error remains relatively high at 4.22 m/s. However, as we sequentially integrate Q, Z, SST, and IR, the model's prediction accuracy improves progressively. Ultimately, when all these climate factors are included, the model achieves a 6.8% reduction in the four-year average error on the test set compared to the baseline model. These results strongly demonstrate the effectiveness of incorporating multiple meteorological data sources in enhancing the accuracy of tropical cyclone intensity forecasting.

TABLE V
ABLATION STUDY OF CLIMATE FACTORS BY MAE FOR TROPICAL CYCLONE INTENSITY FORECASTING FOR LEAD TIMES OF 24 H (UNIT: M/S)

Methods	2015	2016	2017	2018	Avg
UVO	4.15	5.40	3.50	**3.84**	4.22
UVO+Q	4.08	4.90	3.68	4.03	4.18
UVO+Q+Z	**3.91**	4.88	3.62	4.01	4.10
UVO+Q+Z+SST	4.03	4.62	**3.40**	4.10	4.04
UVO+Q+Z+SST+IR	4.00	**4.36**	3.48	3.90	**3.93**

b) Different attention modules: In this experiment, we progressively introduced the CBAM attention module, the temporal attention module, and the attention-cropping module to evaluate their specific impact on model accuracy. To establish a basis for comparison, we first defined a baseline model without any attention modules. Table VI shows that the baseline model achieved a four-year average error of 4.18 m/s on the test set. Next, we introduced the CBAM attention module, which reduced the four-year average error to 4.04 m/s. Subsequently, incorporating the temporal attention module further decreased the error to 3.97 m/s. Finally, adding the attention-cropping module reduced the model's four-year average error on the test set to 3.93 m/s. The prediction error was reduced by 5.9% compared to the baseline model. These experimental results demonstrate that the introduced attention modules have

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF ATTENTION MODULES BY MAE FOR TROPICAL CYCLONE INTENSITY FORECASTING FOR LEAD TIMES OF 24 H (UNIT: M/S)

Methods	2015	2016	2017	2018	Avg
Original	4.27	4.82	3.73	**3.89**	4.18
Original+CBAM	4.13	4.60	3.43	4.00	4.04
Original+CBAM+Temporal Attention	4.15	4.42	**3.40**	3.90	3.97
Original+CBAM+Temporal Attention+ Attention-cropping	**4.00**	**4.36**	3.48	3.90	**3.93**

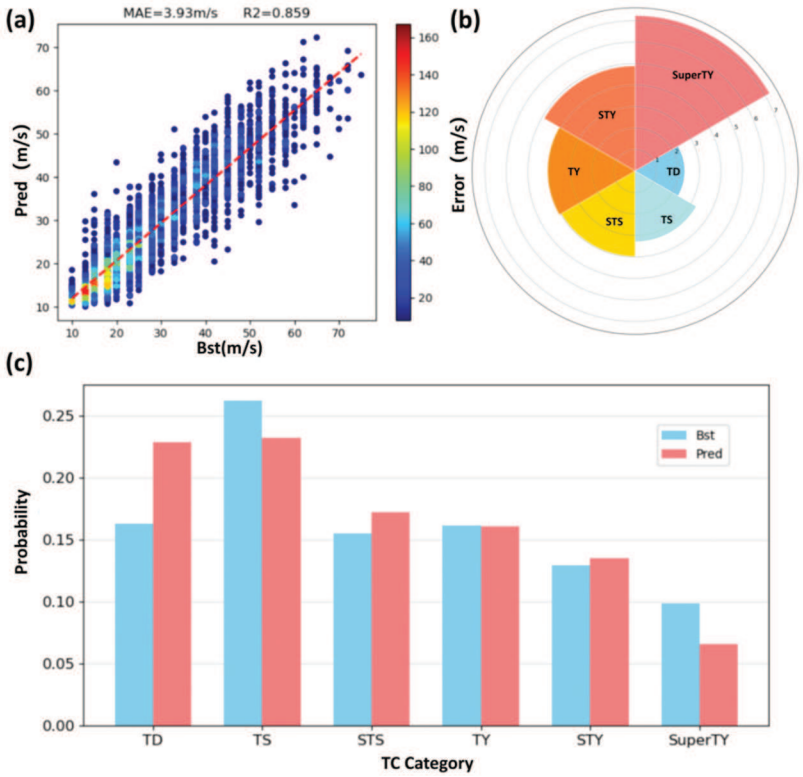


Fig. 5. (a) Scatter plot illustrating the relationship between predicted and best track tropical cyclone intensity using the IRGMF method. (b) Distribution of prediction errors across different tropical cyclone intensity levels for the IRGMF method. (c) Comparison of the frequency distribution of tropical cyclone intensity levels between IRGMF forecasts and best track intensity.

a positive effect on model performance, effectively enhancing both accuracy and overall predictive capability.

D. Discussion

1) Analysis of Forecasting Results: This section analyzes the relationship between the IRGMF's predicted intensity and the best track intensity to further reveal the model's strengths and limitations. Fig. 5(a) presents a scatter plot illustrating the relationship between the predicted intensity and the best track intensity. Each point in the plot represents a tropical cyclone sample, with the x-axis denoting the best track intensity from the CMA-BST dataset and the y-axis representing the model's predicted intensity. The points are closely distributed around a straight line, and the correlation coefficient reaches 0.859, indicating a high consistency between the predicted and the best track intensities. Fig. 5(b) depicts the prediction errors across different intensity levels of tropical cyclones. As shown in the figure, the prediction error increases with the cyclone's intensity level, suggesting that the model performs relatively well for weaker tropical cyclones but exhibits higher errors when predicting stronger ones. Fig. 5(c) compares the frequency distribution of tropical cyclone intensity levels predicted by the IRGMF model with those recorded in the best track dataset. The results indicate that the IRGMF model tends to overpredict the proportion of tropical depressions (TDs) compared to the actual distribution in the best track dataset. Conversely, it underestimates tropical storms (TSs) and super typhoons (SuperTYs). This discrepancy suggests that the model tends to underestimate the intensity of stronger tropical cyclones, misclassifying TS as TD or SuperTY as lower intensity categories. This is attributed to the imbalance in the distribution of tropical cyclone intensity levels within the training samples.

2) Interpretability Analysis of the Attention Module: In this section, we conduct an interpretability analysis of the attention module. We select SuperTY Nangka (the 11th typhoon of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

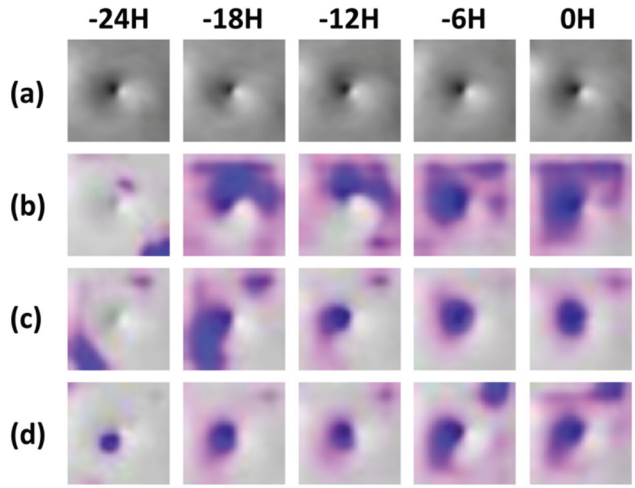


Fig. 6. (a) Original images. (b) Heatmaps generated by the model without attention modules overlaid on the original images. (c) Heatmaps generated by the model with the CBAM attention module overlaid on the original images. (d) Heatmaps generated by the model with both the CBAM and temporal attention modules overlaid on the original images.

2015) from the test set as a case study. By visualizing the heatmaps of V at the 1000-hPa isobaric level, we compare the effects of different attention modules on the model's forecasts. Fig. 6(a) presents the original images, (b) shows the overlays of the heatmaps generated by the model without any attention modules onto the original images, (c) displays the overlay of the heatmaps generated by the model incorporating the CBAM attention module onto the original images, and (d) is an overlay of the heatmaps generated by the model incorporating both the CBAM and temporal attention module onto the original images. Based on Fig. 6, we draw the following conclusions.

Without an attention module [see Fig. 6(b)], the model exhibits a relatively dispersed attention distribution, assigning excessive focus to noncritical pixels, which can introduce unnecessary noise and interfere with prediction accuracy.

After integrating the CBAM attention module [see Fig. 6(c)], the model's attention distribution improves significantly, becoming more concentrated. However, note that when processing sequential data, the model primarily focuses on the central region of the tropical cyclone at the last three time steps (−12, −6, and 0 h), while the attention at the earlier time steps (−24 and −18 h) remains relatively scattered. This suggests that the model does not fully capture the temporal evolution of the tropical cyclone.

After incorporating both the CBAM and temporal attention module into the model [see Fig. 6(d)], we observe that the model effectively focuses its attention on the central region of the tropical cyclone at every time step. The attention distribution becomes more concentrated and exhibits a progressively intensifying trend. This demonstrates that the temporal attention module enables the model to better capture critical time points in the sequential data, thereby enhancing its ability to model the temporal evolution of the tropical cyclone.

Fig. 7 presents the interpretability analysis results of the attention-cropping module. Specifically, Fig. 7(a) displays the infrared satellite images of the tropical cyclone, while Fig. 7(b) shows the corresponding reanalysis data. Fig. 7(c) illustrates the images processed after applying the attention-cropping module.

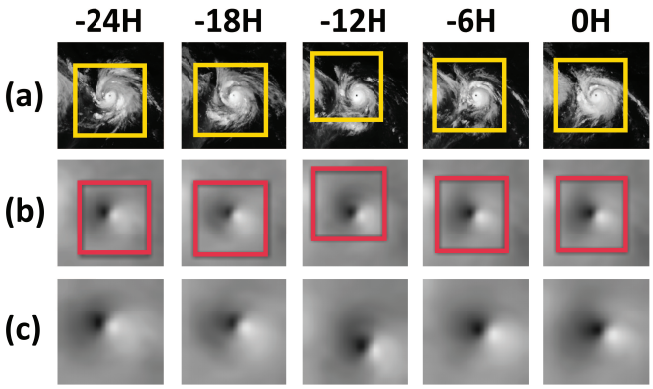


Fig. 7. (a) Infrared satellite images. (b) Reanalysis data. (c) Reanalyze data after cropping.

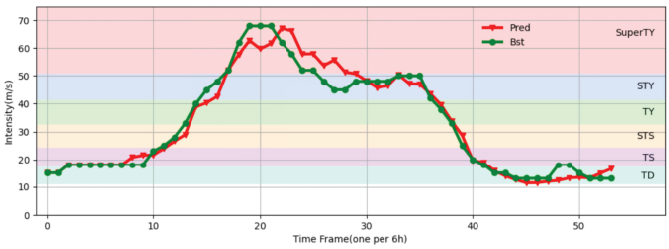


Fig. 8. Comparison of intensity forecasts for Tropical Cyclone Soudelor in 2015.

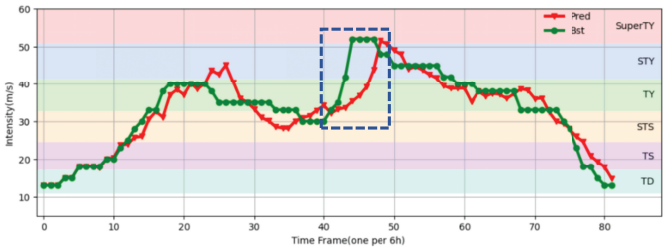


Fig. 9. Comparison of intensity forecasts for Tropical Cyclone NORU in 2017.

The yellow and red boxes in these figures indicate the actual size and position of the cropped image within the infrared satellite images and the reanalysis data, respectively. Observations reveal that the model effectively captures and preserves the core structure and scale characteristics of the tropical cyclone during the cropping process. This significantly reduces the interference of nontarget pixels in the model's analysis and improves the efficiency and accuracy of image processing.

3) Case Study: In this section, we analyze three specific tropical cyclone cases from the test set—SuperTY Soudelor (the 13th typhoon of 2015), SuperTY NORU (the fifth typhoon of 2017), and SuperTY Noul (the sixth typhoon of 2015)—to assess and discuss the effectiveness and limitations of our proposed model in practical applications. We use the model to predict the intensity of these three tropical cyclones and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
PERFORMANCE COMPARISON OF IRGMF BY MAE FOR TROPICAL CYCLONE INTENSITY FORECASTING BETWEEN ABRUPT AND NORMAL INTENSITY CHANGE CASES (UNIT: M/S)

Type	Attribute	2015	2016	2017	2018	Avg
Abrupt Change	Sample Number	71	75	29	61	59
	MAE	8.08	8.90	8.95	10.02	8.98
Normal Change	Sample Number	838	418	518	748	630
	MAE	3.65	3.54	3.17	3.40	3.44

TABLE VIII
PERFORMANCE COMPARISON OF IRGMF BY MAE FOR TROPICAL CYCLONE INTENSITY FORECASTING AT DIFFERENT FORECAST LEAD TIMES (UNIT: M/S)

Lead Times	Methods	2015	2016	2017	2018	Avg
6h	CMA-TRAMS	5.20	4.40	3.20	3.90	4.17
	IRGMF	1.55	1.68	1.41	1.41	1.51
12h	CMA-TRAMS	6.90	6.50	4.20	4.90	5.62
	IRGMF	2.44	2.74	2.06	2.36	2.40
18h	CMA-TRAMS	7.80	7.60	4.70	5.80	6.47
	IRGMF	3.31	3.51	2.63	3.14	3.14
24h	CMA-TRAMS	8.90	8.00	6.50	6.40	7.45
	IRGMF	4.00	4.36	3.48	3.90	3.93
48h	CMA-TRAMS	10.50	11.50	7.70	8.00	9.42
	IRGMF	6.04	7.44	5.85	6.03	6.34
72h	CMA-TRAMS	10.40	11.60	7.80	8.20	9.50
	IRGMF	6.92	10.33	6.78	6.74	7.69

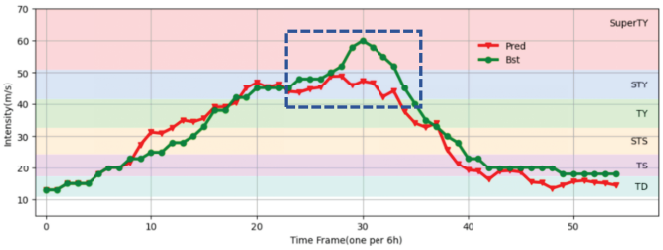


Fig. 10. Comparison of intensity forecasts for Tropical Cyclone Noul in 2015.

compare the predicted results (Pred) with the best track data (Bst) obtained from the CMA-BST dataset. These results are presented as line charts in Figs. 8–10. By examining and analyzing these figures, we derive the following conclusions.

Overall, the predicted results of our model closely follow the trend of the best track data, indicating that our model achieves high accuracy in capturing the general evolution of tropical cyclone intensity, as shown in Fig. 8.

However, as shown by the blue box in Fig. 9, the intensity of Typhoon NORU experiences rapid intensification during its continuous intensification process, and the model's predictive performance is not ideal. Therefore, following the definitions proposed by Kaplan et al. [38] for rapid intensification and by Ma et al. [39] for rapid weakening, we categorized the test samples into two groups: 1) tropical cyclones that exhibited abrupt intensity changes within 24 h and 2) those with normal changes during the same period. As shown in Table VII, the average forecast error of the model over four years was 8.98 m/s for tropical cyclones with abrupt intensity changes, compared to 3.44 m/s for those with normal varying intensity.

In addition, the blue box in Fig. 10 indicates that the model tends to predict slightly lower values. This can be attributed to our dataset's relatively few SuperTY samples, which may have prevented the model from fully learning the characteristics of such strong tropical cyclones.

While our model demonstrates effectiveness in certain aspects, it still requires further enhancement to handle abrupt intensity changes and extreme low-frequency cases better. Future research could focus on addressing these limitations to improve the model's prediction accuracy and applicability.

4) Extended Experiment at Different Forecast Lead Times: To further assess the performance of IRGMF at different forecast lead times, we conducted an extended experiment. Table VIII presents the comparison between IRGMF and CMA-TRAMS at lead times of 6, 12, 18, 24, 48, and 72 h. As shown in the table, IRGMF consistently outperforms CMA-TRAMS, yielding lower prediction errors at all lead times. Moreover, the prediction error of IRGMF increases with longer forecast lead times.

V. CONCLUSION

This article proposes the IRGMF model for forecasting tropical cyclone intensity. IRGMF makes feature representations more comprehensive and richer by integrating more climate factors. We also designed a temporal attention module to guide the model to focus on critical points in sequential data, thereby accurately capturing the dynamic evolution of climate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

factors related to cyclone intensity and improving the model's ability to analyze sequential information. To further optimize accuracy, we introduce an innovative attention-cropping module that precisely crops the reanalysis data of tropical cyclones and eliminates irrelevant information outside the cyclone-scale region. This ensures the accuracy and reliability of the forecasting results. Extensive experimental evaluations demonstrate the superiority and effectiveness of the proposed IRGMF model in tropical cyclone intensity forecasting and provide more accurate information and technical support for meteorological forecasting and decision-making.

However, the proposed model exhibits significant errors when predicting the intensity of rapidly changing tropical cyclones. Thus, we plan to further refine this model to improve its forecasting accuracy for extreme intensity variations.

ACKNOWLEDGMENT

The authors thank LetPub (www.letpub.com.cn) for its linguistic assistance during the preparation of this article.

REFERENCES

[1] R. Chen, W. Zhang, and X. Wang, "Machine learning in tropical cyclone forecast modeling: A review," *Atmosphere*, vol. 11, no. 7, p. 676, Jun. 2020.

[2] J. P. Cangialosi et al., "Recent progress in tropical cyclone intensity forecasting at the national hurricane center," *Weather Forecasting*, vol. 35, no. 5, pp. 1913–1922, Oct. 2020.

[3] H. M. Zahera, M. Ahmed Sherif, and A. Ngonga, "Semantic-based end-to-end learning for typhoon intensity prediction," 2020, *arXiv:2003.13779*.

[4] S. Bao, Z. Zhang, E. Kalina, and B. Liu, "The use of composite GOES-R satellite imagery to evaluate a TC intensity and vortex structure forecast by an FV3GFS-based hurricane forecast model," *Atmosphere*, vol. 13, no. 1, p. 126, Jan. 2022.

[5] P. Wang, P. Wang, C. Wang, B. Xue, and D. Wang, "Using a 3D convolutional neural network and gated recurrent unit for tropical cyclone track forecasting," *Atmos. Res.*, vol. 269, May 2022, Art. no. 106053.

[6] S. Neetu et al., "Global assessment of tropical cyclone intensity statistical–dynamical hindcasts," *Quart. J. Roy. Meteorological Soc.*, vol. 143, no. 706, pp. 2143–2156, Jul. 2017.

[7] J. Yan, L. Mu, L. Wang, R. Ranjan, and A. Y. Zomaya, "Temporal convolutional networks for the advance prediction of ENSO," *Sci. Rep.*, vol. 10, no. 1, p. 8055, May 2020.

[8] K. Janowicz, S. Gao, G. McKenzie, Y. Hu, and B. Bhaduri, "GeoAI: Spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond," *Int. J. Geographical Inf. Sci.*, vol. 34, no. 4, pp. 625–636, Apr. 2020.

[9] L. Boussioux, C. Zeng, T. Guénais, and D. Bertsimas, "Hurricane forecasting: A novel multimodal machine learning framework," *Weather Forecasting*, vol. 37, no. 6, pp. 817–831, Jun. 2022.

[10] Z. Liu, K. Hao, X. Geng, and Z. Shi, "Dual-branched spatio-temporal fusion network for multi-horizon tropical cyclone track forecast," 2022, *arXiv:2202.13336*.

[11] R. Zhang, Q. Liu, R. Hang, and G. Liu, "Predicting tropical cyclogenesis using a deep learning method from gridded satellite and ERA5 reanalysis data in the western North Pacific basin," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, Apr. 2021, Art. no. 4101810.

[12] F. Meng, K. Yang, Y. Yao, Z. Wang, and T. Song, "Tropical cyclone intensity probabilistic forecasting system based on deep learning," *Int. J. Intell. Syst.*, vol. 2023, no. 1, Jan. 2023, Art. no. 3569538.

[13] W. Li, Y. Li, Y. Yang, D. Song, Z. Wei, and A.-A. Liu, "Multimodal adversarial fusion for typhoon intensity forecasting," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4102612.

[14] G. Xu, K. Lin, X. Li, and Y. Ye, "SAF-Net: A spatio-temporal deep learning method for typhoon intensity prediction," *Pattern Recognit. Lett.*, vol. 155, pp. 121–127, Mar. 2022.

[15] C. Wang, X. Li, and G. Zheng, "Tropical cyclone intensity forecasting using model knowledge guided deep learning model," *Environ. Res. Lett.*, vol. 19, no. 2, Jan. 2024, Art. no. 024006.

[16] S. Jiang, H. Fan, and C. Wang, "Improvement of typhoon intensity forecasting by using a novel spatio-temporal deep learning model," *Remote Sens.*, vol. 14, no. 20, p. 5205, Oct. 2022.

[17] D. Ma, L. Wang, S. Fang, and J. Lin, "Tropical cyclone intensity prediction by inter- and intra-pattern fusion based on multi-source data," *Environ. Res. Lett.*, vol. 18, no. 1, Jan. 2023, Art. no. 014020.

[18] X. Yang et al., "Verification of GRAPES unified global and regional numerical weather prediction model dynamic core," *Chin. Sci. Bull.*, vol. 53, no. 22, pp. 3458–3464, Nov. 2008.

[19] J.-J. Morcrette, A. Beljaars, A. Benedetti, L. Jones, and O. Boucher, "Sea-salt and dust aerosols in the ECMWF IFS model," *Geophys. Res. Lett.*, vol. 35, no. 24, Dec. 2008, Art. no. L24813.

[20] S. Prakash, I. Momin, A. Mitra, P. S. Bhattacharjee, F. Yang, and V. Tallapragada, "An early assessment of medium range monsoon precipitation forecasts from the latest high-resolution NCEP-GFS (T1534) model over South Asia," *Pure Appl. Geophys.*, vol. 173, no. 6, pp. 2215–2225, Jun. 2016.

[21] B. R. Jarvinen and C. J. Neumann, "Statistical forecasts of tropical cyclone intensity for the North Atlantic basin," NOAA, Tech. Memo. NWS NHC-10, Apr. 1979, p.22.

[22] J. H. Chu, "A regression model for the western North Pacific tropical cyclone intensity forecast," NRL, Memo. Rep., Nov. 1994, vol. 7541-94-7215, p.33.

[23] M. DeMaria and J. Kaplan, "A statistical hurricane intensity prediction scheme (SHIPS) for the Atlantic basin," *Weather Forecasting*, vol. 9, no. 2, pp. 209–220, Mar. 1994.

[24] W. Zhang, G. Villarini, G. A. Vecchi, H. Murakami, and R. Gudgel, "Statistical-dynamical seasonal forecast of western North Pacific and East Asia landfalling tropical cyclones using the high-resolution GFDL FLOR coupled model," *J. Adv. Model. Earth Syst.*, vol. 8, no. 2, pp. 538–565, Jun. 2016.

[25] B. Tong, X. Wang, J. Y. Fu, P. W. Chan, and Y. C. He, "Short-term prediction of the intensity and track of tropical cyclone via ConvLSTM model," *J. Wind Eng. Ind. Aerodynamics*, vol. 226, Jul. 2022, Art. no. 105026.

[26] C. Huang, C. Bai, S. Chan, and J. Zhang, "MMSTN: A multi-modal spatial–temporal network for tropical cyclone short-term prediction," *Geophys. Res. Lett.*, vol. 49, no. 4, Feb. 2022, Art. no. e2021GL096898.

[27] C. Huang, C. Bai, S. Chan, J. Zhang, and Y. Wu, "MGTCF: Multi-generator tropical cyclone forecasting with heterogeneous meteorological data," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2023, vol. 37, no. 4, pp. 5096–5104.

[28] Y.-J. Park et al., "Long-term typhoon trajectory prediction: A physics-conditioned approach without reanalysis data," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Apr. 2024, pp. 1–11.

[29] Q. Jin, X. Fan, J. Liu, Z. Xue, and H. Jian, "Using eXtreme gradient BOOSTing to predict changes in tropical cyclone intensity over the western North Pacific," *Atmosphere*, vol. 10, no. 6, p. 341, Jun. 2019.

[30] J. A. Knaff, C. R. Sampson, and M. DeMaria, "An operational statistical typhoon intensity prediction scheme for the western North Pacific," *Weather Forecasting*, vol. 20, no. 4, pp. 688–699, Aug. 2005.

[31] M. Ying et al., "An overview of the China meteorological administration tropical cyclone database," *J. Atmos. Ocean. Technol.*, vol. 31, no. 2, pp. 287–301, Feb. 2014.

[32] H. Hersbach et al., "ERA5 hourly data on pressure levels from 1940 to present," Copernicus Climate Change Service (C3S) Climate Data Store (CDS), 2023, doi: 10.24381/cds.bd0915c6.

[33] X. Wu et al., "Stratospheric gravity waves as a proxy for hurricane intensification: A case study of weather research and forecast simulation for hurricane Joaquin," *Geophys. Res. Lett.*, vol. 49, no. 1, Jan. 2022, Art. no. e2021GL097010.

[34] K. A. Hill and G. M. Lackmann, "Influence of environmental humidity on tropical cyclone size," *Monthly Weather Rev.*, vol. 137, no. 10, pp. 3294–3315, Oct. 2009.

[35] M. Mandal, U. C. Mohanty, P. Sinha, and M. M. Ali, "Impact of sea surface temperature in modulating movement and intensity of tropical cyclones," *Natural Hazards*, vol. 41, no. 3, pp. 413–427, Jun. 2007.

[36] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.

[37] K. R. Knapp et al., "Globally gridded satellite observations for climate studies," *Bull. Amer. Meteorological Soc.*, vol. 92, no. 7, pp. 893–907, Jul. 2011.

[38] J. Kaplan, M. DeMaria, and J. A. Knaff, "A revised tropical cyclone rapid intensification index for the Atlantic and eastern North Pacific basins," *Weather Forecasting*, vol. 25, no. 1, pp. 220–241, Feb. 2010.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.

[39] Z. Ma, J. Fei, and X. Huang, "A definition of rapid weakening for tropical cyclones over the western North Pacific," *Geophys. Res. Lett.*, vol. 46, no. 20, pp. 11471–11478, Oct. 2019.
[40] G. Chen, L. Bai, and R. Wan, "Verification on forecasts of tropical cyclones over western North Pacific in 2015," *Meteorological Monthly*, vol. 43, no. 4, pp. 501–507, Apr. 2017.
[41] G. Chen, X. Zhang, L. Bai, and R. Wan, "Verification on forecasts of tropical cyclones over western North Pacific in 2016," *Meteorological Monthly*, vol. 44, no. 4, pp. 582–589, Apr. 2018.
[42] G. Chen, X. Zhang, L. Bai, and R. Wan, "Verification on forecasts of tropical cyclones over western North Pacific and South China sea in 2017," *Meteorological Monthly*, vol. 45, no. 4, pp. 577–586, Apr. 2019.
[43] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 785–794.
[44] C. Wang, G. Zheng, X. Li, Q. Xu, B. Liu, and J. Zhang, "Tropical cyclone intensity estimation from geostationary satellite imagery using deep convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Mar. 2022, Art. no. 4101416.



Chang-Jiang Zhang was born in Heilongjiang, China, in 1974. He received the Ph.D. degree in control theory and control engineering from Beijing Institute of Technology, Beijing, China, in 2004.

He was the Middle-Aged Discipline Leader of Zhejiang Normal University, Jinhua, China, in 2010. From 2012 to 2013, he was a Research Fellow at Cornell University, Ithaca, NY, USA. From 2009 to 2021, he was a Professor at the Electronic and Information Engineering Department, College of Physics and Electronic Information, Zhejiang Normal University. He is currently a Professor with the Electronic and Information Engineering Department, School of Artificial Intelligence, Taizhou University, Taizhou, China, where he is the Head of the Laboratory of Intelligent Vision and Meteorological Big Data, Institute of Intelligent Perception and Cross Application. He is the author of three books, more than 100 articles, and more than 20 inventions. He is the principal investigator of more than 20 foundations and projects. His research interests include image processing, machine learning, multiscale geometry analysis, medical informatics, and applied meteorology.

Dr. Zhang's research achievements were awarded First Prize, Second Prize, and Third Prize by the Education Department of Zhejiang Province, China. He was awarded the Second Level of the New Century 151 Talent Project of Zhejiang Province, China.



Li-Yu Xu was born in Zhejiang, China, in 2000. He is currently pursuing the M.S. degree with the College of Physics and Electronic Information Engineering, Zhejiang Normal University, Jinhua, Zhejiang, China.

His research interests include image processing and deep learning.



Lei-Ming Ma was born in China, in 1975. He received the Ph.D. degree in meteorology from Nanjing University, Nanjing, China, in 2009.

He is currently a Research Fellow and a Professor at Shanghai Typhoon Institute, China Meteorological Administration, Shanghai, China. He is a Chief Artificial Intelligence Officer at Shanghai Meteorological Bureau, Shanghai. He is the author of more than 20 articles. He is a principal investigator of more than ten foundations and projects. His research interests include numerical forecasting theory and methods for tropical cyclones.

Dr. Ma is a Committee Member of the Numerical Forecasting Committee and China Meteorological Society. His awards and honors include the Science and Technology Progress Award of Shanghai City, the National Ocean Innovation Award of China, and the National Outstanding Young Meteorologist of China.



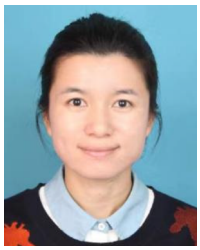
Xiao-Qin Lu was born in China, in 1975. She received the M.S. degree in meteorology from East China Normal University, Shanghai, China, in 2003.

She is currently a Research Fellow with Shanghai Typhoon Institute, China Meteorological Administration, Shanghai. She mainly engages in research on objective analysis technology of tropical cyclone structure and the application of tropical cyclone multisource data mining.



Peng He was born in China, in 1970. He received the B.S. degree in semiconductor physics and devices from Harbin Institute of Technology, Harbin, China, in 1991, the M.S. degree in computer application from Qiqihar University, Qiqihar, China, in 2003, and the Ph.D. degree in precision instruments and machinery from Harbin University of Science and Technology, Harbin, in 2013.

He is currently a Professor with the School of Artificial Intelligence, Taizhou University, Taizhou, China. He has published over 100 papers in journals and conferences. His research mainly focuses on embedded product design, artificial intelligence algorithms, machine vision, and digital image processing.



Xue Wei was born in China, in 1985. She received the M.S. degree in atmospheric remote sensing science and technology from Nanjing University of Information Science and Technology, Nanjing, China, in 2011.

She is currently an Engineer with Taizhou Meteorological Bureau, Taizhou, Zhejiang, China, where she is the main Member of the Professional Meteorological Service Team and the Research Team for applying strong convective forecast technology. She mainly engages in research on precise meteorological service methods, analysis, and prediction of severe convective weather.



Tao Wang was born in China, in 1979. He received the M.S. degree in computer technology engineering from Zhejiang University of Technology, Hangzhou, China, in 2014.

He is currently a Lecturer with the School of Artificial Intelligence, Taizhou University, Taizhou, China. He mainly engages in research on intelligent control and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:53:02 UTC from IEEE Xplore. Restrictions apply.